

JIHAN CHOWDHURY

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EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Engineering in Computer Engineering (Software Option), GPA: 3.77

Expected Apr. 2027

Relevant Coursework: Algorithms & Data Structures, Database Systems, Operating Systems, Object-Oriented Analysis & Design, Software Architecture

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript/TypeScript, HTML/CSS, SQL, PHP

Frameworks: FastAPI, Flask, Next.js, React, Node.js, Express, TailwindCSS

Databases: PostgreSQL, MongoDB, Oracle SQL

Technologies: Git/GitHub, Docker, Linux/Unix, AWS, REST APIs, Azure, Vercel, Railway

EXPERIENCE

Hitachi Rail

Feb. 2026 – Present

AI Intern – Procurement

Toronto, ON

- Built a **Python-based quotation and price-list extraction pipeline** for heterogeneous supplier files including **PDF, XLSX, and MSG**, using an **Azure AI endpoint** to transform unstructured documents into structured pricing data.
- Extracted commercial fields including **material numbers, manufacturer part numbers, pricing, currency, units of measure, quote dates, and validity dates** into standardized Excel datasets used for **SAP pricing and bid-costing workflows**.
- Engineered an **AI-assisted PDF redaction and editing pipeline** for hundreds of non-searchable documents after direct AI-based PDF editing proved unreliable; combined **EasyOCR**, page images, and word-level coordinate metadata to identify and programmatically modify exact document content.
- Developing a **Python-based invoice coding automation** with procurement stakeholders, targeting an estimated **20–30 hours/month** reduction in manual processing and integration with enterprise workflows through **Power Automate**.

PROJECTS

League Insights AI | Python, Flask, PostgreSQL, AWS, Next.js

Nov. 2025 – Present

- Led technical execution of a full-stack analytics platform, owning an **asynchronous Flask API (Python)** with **PostgreSQL** serving **200+ distinct players in a month** and supporting **1,000+ match records per player**.
- Implemented **SQL-backed data models** and integrity checks to support **large-scale aggregation**.
- Integrated **AWS-hosted AI report generation** and exposed endpoints consumed by a **Next.js dashboard**.
- Directed technical execution across a **3-person team** by defining backend interfaces and task breakdowns.

E-Library Management Platform | Next.js, TypeScript, PostgreSQL, Drizzle ORM, NextAuth.js

Oct. 2025

- Built a **library management system** with dashboards, real-time availability, and due-date logic.
- Implemented **role-based authentication** using NextAuth and **type-safe database access** with Drizzle ORM.
- Optimized searches with debounced filtering, reducing redundant search-related queries by **75%**.
- Deployed the application on Vercel using its GitHub integration for automatic builds and deployments.

E-Commerce DBMS | Oracle SQL, PHP, Unix

Sep. 2025 – Nov. 2025

- Designed a **normalized relational e-commerce database** (11 tables, 3NF/BCNF) to ensure data integrity and transactional consistency.
- Implemented **advanced SQL analytics** using multi-table JOINS, aggregations, and subqueries to support reporting and order workflows.
- Automated schema deployment and data seeding through **Unix-based scripting**, enabling repeatable and reliable database setup.